

Category	Commands
Navigation	<code>pwd</code> , <code>ls</code> , <code>cd</code>
Files	<code>mkdir</code> , <code>touch</code> , <code>cp</code> , <code>mv</code> , <code>rm</code> , <code>cat</code> , <code>find</code>
Text Processing	<code>grep</code> , <code>awk</code> , <code>sed</code> , <code>sort</code> , <code>uniq</code> , <code>cut</code> , <code>wc</code> , <code>head</code> , <code>tail</code>
Permission	<code>chmod</code> , <code>chown</code> , <code>umask</code> , <code>sudo</code>
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Show more 	

Below is a practical Linux command reference with **3 examples for each command**, including **what each command does** and **what it achieves**. This is suitable for B.Tech students learning Linux administration.

1. Navigation Commands

`pwd` (Print Working Directory)

Displays the current directory.

Example	Description	Achieves
<code>pwd</code>	Show current location	Displays <code>/home/student</code>
<code>cd /etc && pwd</code>	Move to <code>/etc</code> and print location	Confirms current directory is <code>/etc</code>
<code>mkdir test && cd test && pwd</code>	Create directory and enter it	Shows full path of new folder

ls (List Files)

Example	Description	Achieves
ls	List files	Shows file names
ls -l	Long listing	Shows permissions, owner, size
ls -la	Include hidden files	Displays hidden files beginning with .

cd (Change Directory)

Example	Description	Achieves
cd /home/student	Move to directory	Opens specified directory
cd ..	Move one level up	Goes to parent directory
cd ~	Go to home directory	Returns to user's home

2. File Commands

mkdir

Example	Description	Achieves
mkdir project	Create folder	Makes directory project
mkdir dir1 dir2	Create multiple folders	Creates two directories
mkdir -p a/b/c	Create nested folders	Creates entire directory tree

touch

Example	Description	Achieves
<code>touch file.txt</code>	Create empty file	Creates new file
<code>touch f1 f2 f3</code>	Create multiple files	Creates three files
<code>touch *.log</code>	Update timestamps	Changes modification time

cp

Example	Description	Achieves
<code>cp file1 file2</code>	Copy file	Creates duplicate
<code>cp file.txt backup/</code>	Copy into folder	Places copy in backup
<code>cp -r dir1 dir2</code>	Copy directory	Copies entire directory

mv

Example	Description	Achieves
<code>mv file1 file2</code>	Rename file	Changes filename
<code>mv file.txt docs/</code>	Move file	Moves file into folder
<code>mv *.txt archive/</code>	Move all text files	Organizes files

rm

Example	Description	Achieves
<code>rm file.txt</code>	Delete file	Removes file
<code>rm -r project</code>	Delete directory	Removes folder recursively

Example	Description	Achieves
<code>rm *.tmp</code>	Delete temp files	Cleans temporary files

cat

Example	Description	Achieves
<code>cat file.txt</code>	Display contents	Reads file
<code>cat f1 f2</code>	Combine files	Prints both files
<code>cat > notes.txt</code>	Create file from keyboard	Saves typed input

find

Example	Description	Achieves
<code>find . -name "*.txt"</code>	Find text files	Searches current directory
<code>find /home -type d</code>	Find directories	Lists directories
<code>find . -size +10M</code>	Find large files	Lists files over 10 MB

3. Text Processing

grep

Example	Description	Achieves
<code>grep "error" log.txt</code>	Search word	Finds matching lines
<code>grep -i linux notes.txt</code>	Ignore case	Finds Linux/linux
<code>grep -r "main" src/</code>	Recursive search	Searches source files

awk

Example	Description	Achieves
awk '{print \$1}' file.txt	Print first column	Displays first field
awk '{sum+=\$2} END{print sum}' marks.txt	Sum column	Calculates total
awk '/error/{print}' log.txt	Print matching lines	Filters records

sed

Example	Description	Achieves
sed 's/apple/orange/' file.txt	Replace first occurrence	Text substitution
sed 's/apple/orange/g' file.txt	Replace all	Global replacement
sed -n '1,5p' file.txt	Print first five lines	Displays selected lines

sort

Example	Description	Achieves
sort names.txt	Alphabetical sort	Orders lines
sort -r names.txt	Reverse order	Descending sort
sort -n marks.txt	Numeric sort	Orders numbers

uniq

Example	Description	Achieves
<code>uniq file.txt</code>	Remove duplicates	Shows unique lines
<code>uniq -c file.txt</code>	Count duplicates	Frequency count
<code>sort file.txt uniq</code>	Unique sorted lines	Removes duplicates

cut

Example	Description	Achieves
<code>cut -d',' -f1 data.csv</code>	Extract first column	Displays first field
<code>cut -c1-5 file.txt</code>	First five characters	Character extraction
<code>cut -f2 marks.txt</code>	Second tab field	Displays second column

WC

Example	Description	Achieves
<code>wc file.txt</code>	Count lines, words, bytes	File statistics
<code>wc -l file.txt</code>	Count lines	Number of lines
<code>wc -w file.txt</code>	Count words	Number of words

head

Example	Description	Achieves
<code>head file.txt</code>	First 10 lines	Preview file
<code>head -5 file.txt</code>	First five lines	Displays first five

Example	Description	Achieves
<code>head *.log</code>	First lines of logs	Quick inspection

tail

Example	Description	Achieves
<code>tail file.txt</code>	Last 10 lines	View file end
<code>tail -5 file.txt</code>	Last five lines	Recent records
<code>tail -f logfile.log</code>	Live monitoring	Watches log updates

4. Permissions

chmod

Example	Description	Achieves
<code>chmod 755 script.sh</code>	Set permissions	Owner full access
<code>chmod +x run.sh</code>	Make executable	Allows execution
<code>chmod -R 777 data</code>	Recursive permissions	Applies to all files

chown

Example	Description	Achieves
<code>sudo chown user file.txt</code>	Change owner	New file owner
<code>sudo chown user:group file.txt</code>	Change owner/group	Updates ownership
<code>sudo chown -R user dir</code>	Recursive ownership	Changes entire folder

umask

Example	Description	Achieves
<code>umask</code>	View current mask	Shows default permissions
<code>umask 022</code>	Set default permissions	Files become 644
<code>umask 077</code>	Restrictive permissions	Only owner has access

sudo

Example	Description	Achieves
<code>sudo apt update</code>	Run as root	Updates package list
<code>sudo reboot</code>	Restart system	Reboots computer
<code>sudo useradd john</code>	Create user	Administrative task

5. User Management

useradd

- `sudo useradd alice` → Creates user `alice`
- `sudo useradd -m bob` → Creates user with home directory
- `sudo useradd -s /bin/bash dev` → Creates user with Bash shell

usermod

- `sudo usermod -aG sudo alice` → Adds user to `sudo` group
- `sudo usermod -l newname oldname` → Renames user
- `sudo usermod -d /home/newhome alice` → Changes home directory

userdel

- `sudo userdel alice` → Deletes user
- `sudo userdel -r alice` → Deletes user and home directory

- `sudo userdel tempuser` → Removes temporary account

passwd

- `passwd` → Change your password
- `sudo passwd alice` → Set password for `alice`
- `sudo passwd -l alice` → Lock user account

groups

- `groups` → Show current user's groups
- `groups alice` → Show Alice's groups
- `groups root` → Show root's groups

id

- `id` → Show current UID/GID
 - `id alice` → Show Alice's UID/GID
 - `id -u` → Show only UID
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6. Process Management

ps

- `ps` → Running processes
- `ps -ef` → All processes
- `ps aux` → Detailed process list

top

- `top` → Live process monitor
- `top -u student` → Processes for one user
- `top -d 5` → Refresh every 5 seconds

kill

- `kill 1234` → Terminate process
- `kill -9 1234` → Force terminate
- `kill -15 1234` → Graceful termination

killall

- `killall firefox` → Kill all Firefox processes
- `killall python` → Stop all Python programs
- `killall -9 java` → Force stop Java processes

systemctl

- `systemctl status ssh` → Service status
 - `sudo systemctl start apache2` → Start service
 - `sudo systemctl enable nginx` → Enable at boot
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7. System Monitoring

free

- `free` → Memory usage
- `free -h` → Human-readable memory
- `free -m` → Memory in MB

df

- `df -h` → Disk usage
- `df -T` → Show filesystem type
- `df /home` → Disk usage of /home

du

- `du -sh folder` → Folder size
- `du -ah` → All file sizes
- `du --max-depth=1` → Size by subdirectory

vmstat

- `vmstat` → Virtual memory statistics
- `vmstat 2` → Update every 2 seconds
- `vmstat 1 5` → Five updates at 1-second intervals

iostat

- `iostat` → Disk I/O statistics
- `iostat -x` → Extended disk statistics
- `iostat 2` → Update every 2 seconds

journalctl

- `journalctl` → View system logs
 - `journalctl -u ssh` → SSH service logs
 - `journalctl -f` → Follow logs in real time
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8. Networking

ping

- `ping google.com` → Check connectivity
- `ping -c 4 google.com` → Send four packets
- `ping 8.8.8.8` → Test Internet access

ss

- `ss -t` → TCP connections
- `ss -l` → Listening ports
- `ss -tunlp` → Detailed socket information

netstat

- `netstat -an` → All connections
- `netstat -r` → Routing table
- `netstat -tulpn` → Listening services

traceroute

- `traceroute google.com` → Show network path
- `traceroute 8.8.8.8` → Trace by IP
- `traceroute -m 10 google.com` → Limit hops

ssh

- `ssh user@192.168.1.100` → Remote login
 - `ssh -p 2222 user@server` → Connect using custom port
 - `ssh user@host "ls -l"` → Execute remote command
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9. Shell Scripting

Variables

- `name="John"` → Store text
- `age=20` → Store number
- `echo $name` → Display variable value

if

- Check if number is positive
- Check if file exists (`[ -f file ]`)
- Check if user is root (`$EUID -eq 0`)

for

- `for i in {1..5}` → Loop five times
- `for file in *.txt` → Process text files
- `for user in alice bob` → Iterate through names

while

- Countdown loop
- Read file line by line
- Wait until a condition changes

Functions

- `greet(){ echo "Hello"; }` → Reusable greeting
- `add(){ echo $(( $1+$2 )); }` → Add two numbers
- `backup(){ tar -czf backup.tar.gz $1; }` → Backup function

crontab

- `crontab -l` → List scheduled jobs
- `crontab -e` → Edit cron jobs
- `0 2 * * * backup.sh` → Run backup daily at 2 AM

10. Software Management

apt

- `sudo apt update` → Update package list
- `sudo apt install git` → Install Git

- `sudo apt remove vim` → Remove Vim

yum

- `sudo yum update` → Update packages
 - `sudo yum install httpd` → Install Apache
 - `sudo yum remove httpd` → Remove Apache
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11. Git Commands

git init

- `git init` → Initialize repository
- `git init myproject` → Create new repository
- `git init --bare repo.git` → Create bare repository

git clone

- `git clone <repository-url>` → Clone repository
- `git clone <repository-url> project` → Clone with custom folder name
- `git clone --depth 1 <repository-url>` → Shallow clone

git status

- `git status` → Repository status
- `git status -s` → Short status
- `git status --branch` → Status with branch

git add

- `git add file.txt` → Stage one file
- `git add .` → Stage all changes
- `git add *.py` → Stage Python files

git commit

- `git commit -m "Initial commit"` → Save staged changes
- `git commit -am "Update"` → Stage tracked files and commit
- `git commit --amend` → Modify last commit

git push

- `git push origin main` → Push to remote
- `git push -u origin main` → Set upstream branch
- `git push --all` → Push all branches

`git pull`

- `git pull` → Fetch and merge
- `git pull origin main` → Pull main branch
- `git pull --rebase` → Rebase instead of merge

`git branch`

- `git branch` → List branches
- `git branch feature` → Create branch
- `git branch -d feature` → Delete merged branch

`git log`

- `git log` → Commit history
- `git log --oneline` → Compact history
- `git log --graph --oneline --all` → Visual branch history

12. Backup Commands

`tar`

- `tar -cvf backup.tar folder/` → Create archive
- `tar -xvf backup.tar` → Extract archive
- `tar -czvf backup.tar.gz folder/` → Create compressed archive

`rsync`

- `rsync -av source/ backup/` → Synchronize directories
- `rsync -av --delete source/ backup/` → Mirror source exactly
- `rsync -av file.txt user@server:/backup/` → Copy file to a remote server

This collection covers the most common Linux administration commands with practical examples that students can execute in a lab environment.

ChatGPT can make mistakes. Check important info.